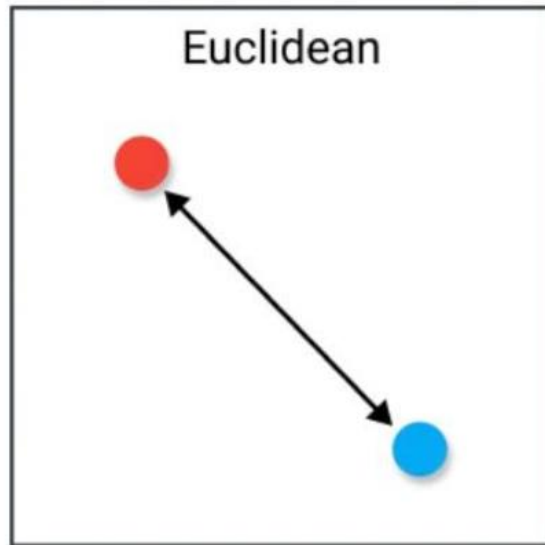


Nature Inspired Algorithms

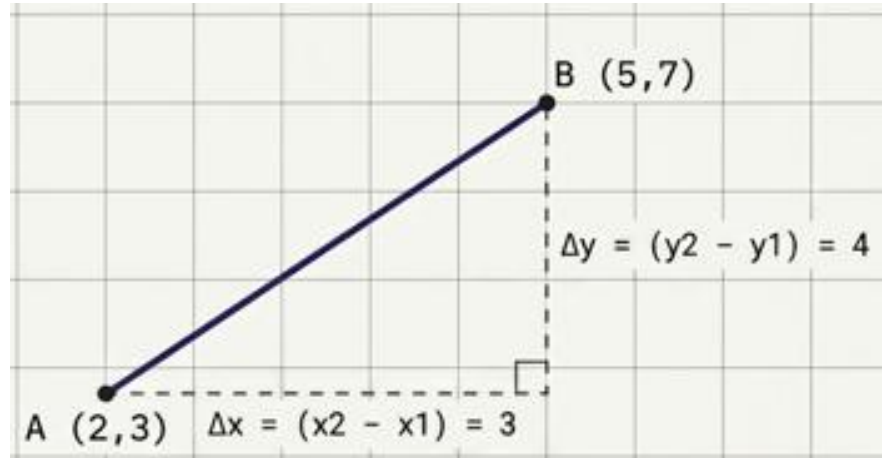
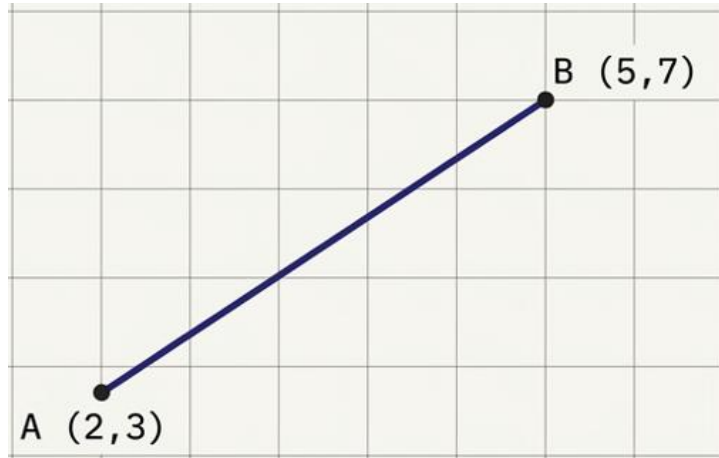
Distance and Similarity Measures

Euclidean Distance

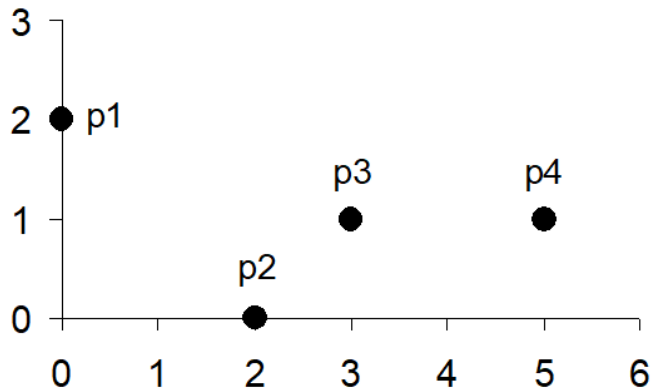
$$D(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$



Euclidean Distance



Euclidean Distance



point	x	y
p1	0	2
p2	2	0
p3	3	1
p4	5	1

	p1	p2	p3	p4
p1	0	2.828	3.162	5.099
p2	2.828	0	1.414	3.162
p3	3.162	1.414	0	2
p4	5.099	3.162	2	0

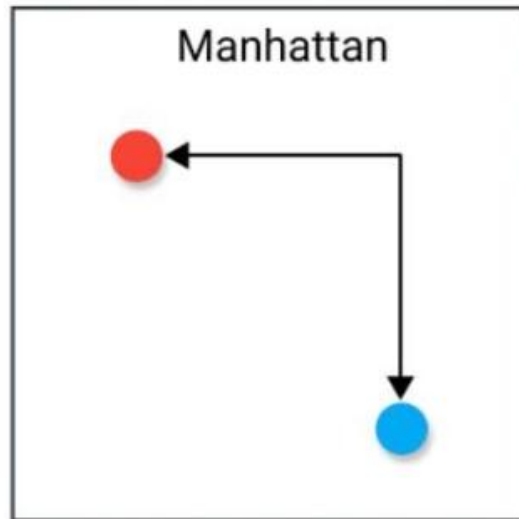
Distance Matrix

Activity

1. In a feature space, two data points are given as $A(2,3)$ and $B(8,11)$. Using the Euclidean distance metric, compute the **distance** between these two points.
2. In a feature space, two data points are given as $A(-2,3)$ and $B(4,-5)$. Using the Euclidean distance metric, compute the **distance** between these two points.
3. In a feature space, three data points are given as $A(1,2,3)$, $B(4,0,5)$, and $C(-2,3,1)$. Using the Euclidean distance metric, compute the **distance matrix** for the given points, where each entry represents the pairwise distance between the corresponding points.

Manhattan Distance

$$D(x, y) = \sum_{i=1}^k |x_i - y_i|$$

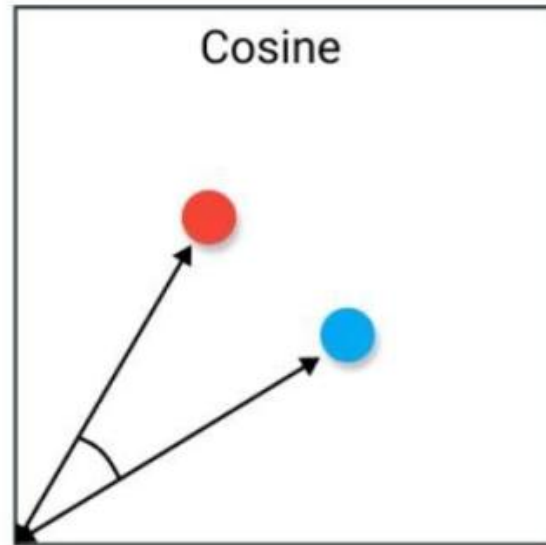


Activity

1. In a feature space, two data points are given as $A(3,-2)$ and $B(7,4)$. Using the Manhattan distance metric, compute the **distance** between the two points.
2. In a feature space, two data points are given as $A(1,2,3)$ and $B(4,-1,5)$. Using the Manhattan distance metric, compute the **distance** between the two points.
3. In a feature space, three data points are given as $A(1,2)$, $B(4,0)$, and $C(-2,3)$. Using the Manhattan distance metric, compute the **distance matrix** for the given points.

Cosine Similarity

$$D(x, y) = \cos(\theta) = \frac{x \cdot y}{\|x\| \|y\|}$$



- Cosine similarity measures the angle-based similarity between two vectors, capturing how aligned their directions are, independent of magnitude.

Cosine Similarity

Cosine similarity = 1

- Same direction
- Very similar patterns

Cosine similarity = 0

- No relationship
- Orthogonal vectors

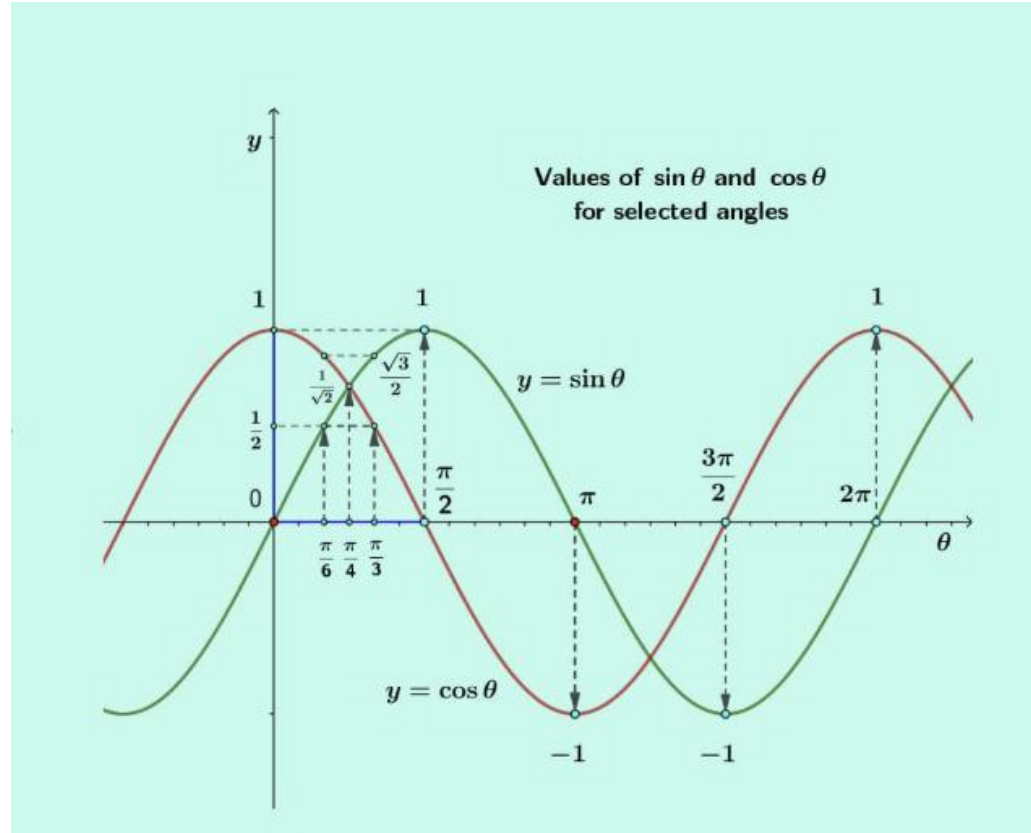
Cosine similarity = -1

- Opposite directions
- Completely different

$$\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \phi$$

Or

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y$$



Activity

1. In a feature space, two vectors are given as $A(1,2)$ and $B(2,3)$. Using the cosine similarity metric, compute the **similarity** between the two vectors and interpret the result in terms of directional similarity.
2. In a feature space, two vectors are given as $A(1,0,1)$ and $B(2,1,3)$. Using cosine similarity, determine how **similar** the two vectors are.
3. In a feature space, three vectors are given as $A(1,2,3)$, $B(4,0,5)$, and $C(2,1,1)$. Using the cosine similarity metric, compute the **cosine similarity matrix** for the given vectors.

Thank you!

